From Replica to Reference: Active Learning in Engineering Education for Digital Twin–Guided Real-System Evolution

João Mello da Silva[1], Mateus Halbe Torres[1], Edison Ishikawa[1], Ari Melo Mariano[1], César Antônio Ciuffo Moreira[2]

[1] University of Brasília-UnB, Brazil
[2] APEX Brasil, Brazil

DIGITAL TWINS AS SOCIO-TECHNICAL SYSTEMS AND IT/OT/ET CONVERGENCE

THE DIGITAL TWIN CONSORTIUM (DTC) DEFINES DIGITAL TWIN AS "A VIRTUAL REPRESENTATION OF REAL-WORLD ENTITIES AND PROCESSES SYNCHRONIZED AT A SPECIFIED FREQUENCY AND FIDELITY, USING REAL-TIME AND HISTORICAL DATA TO REPRESENT THE PAST AND PRESENT AND TO SIMULATE PREDICTED FUTURES".

IN THIS FRAMING, A DIGITAL TWIN IS NOT SIMPLY A MODEL OR DASHBOARD; IT IS A SOCIO-TECHNICAL SYSTEM WHOSE VALUE EMERGES FROM HOW IT SUPPORTS DECISIONS AND INTERVENTIONS IN REAL-LIFE OPERATIONS.

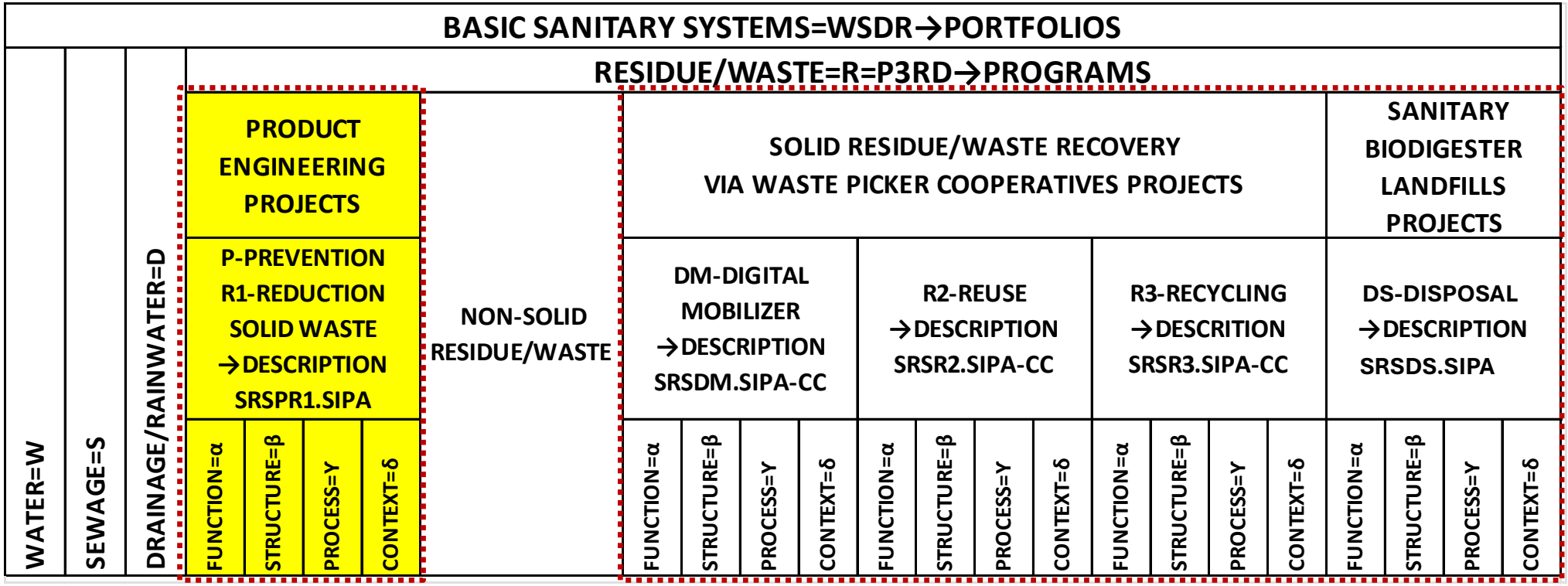
DTC'S EMPHASIS ON IT/OT/ET CONVERGENCE PROVIDES A USEFUL EDUCATIONAL LENS:

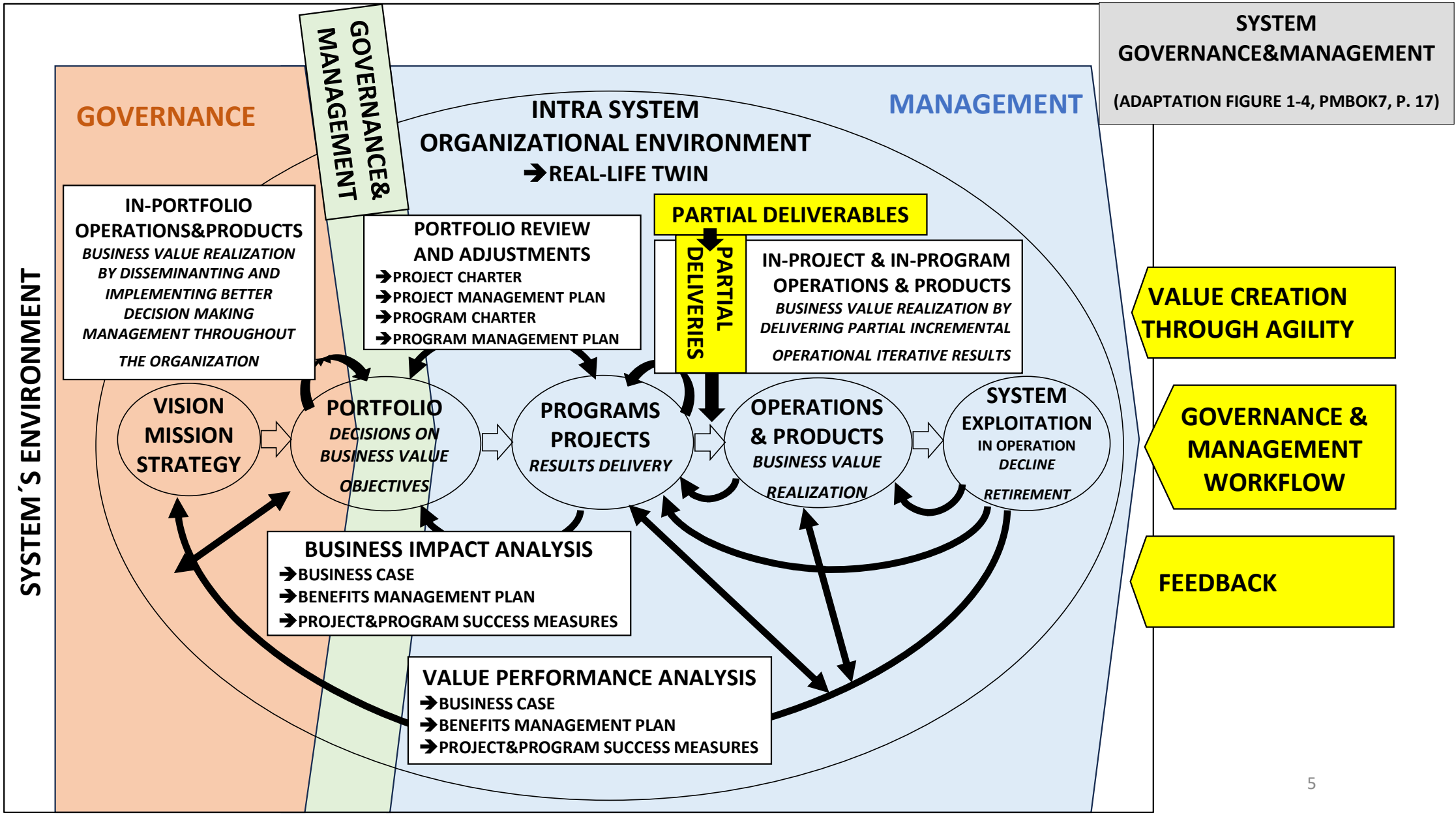
- **INFORMATION TECHNOLOGY→IT (DATA COLLECTION, PROCESSING, STORAGE, COMMUNICATION);**
- **OPERATIONAL TECHNOLOGY→OT (MONITORING/CONTROL OF ESSENTIALLY PHYSICAL PROCESSES); AND**
- **ENGINEERING TECHNOLOGY→ET (MODELING AND SYSTEM DESIGN).**

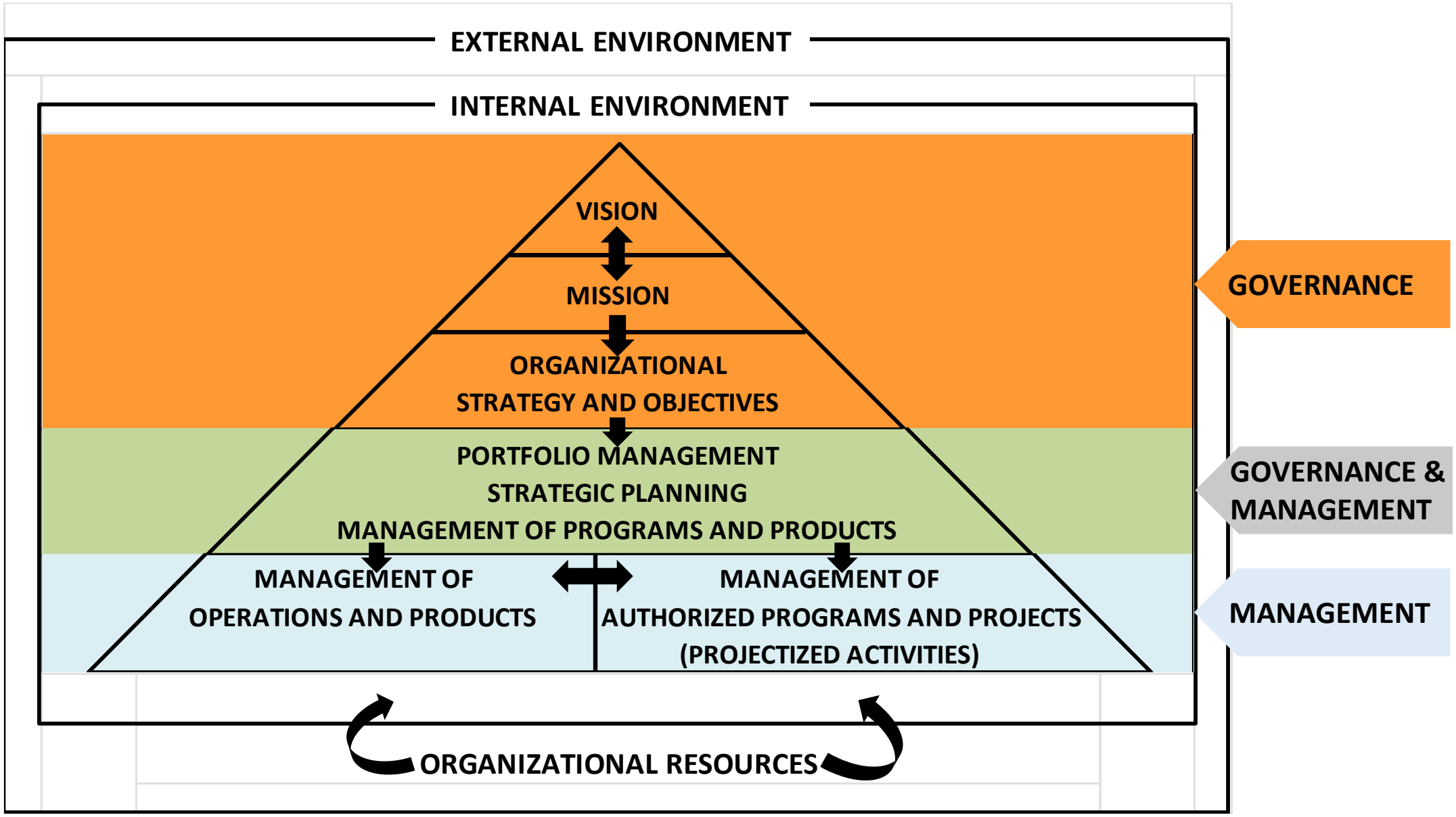
MUST BE INTEGRATED INTO COHERENT INTERVENTION.

DIGITAL TWINS AS SOCIO-TECHNICAL SYSTEMS AND IT/OT/ET CONVERGENCE

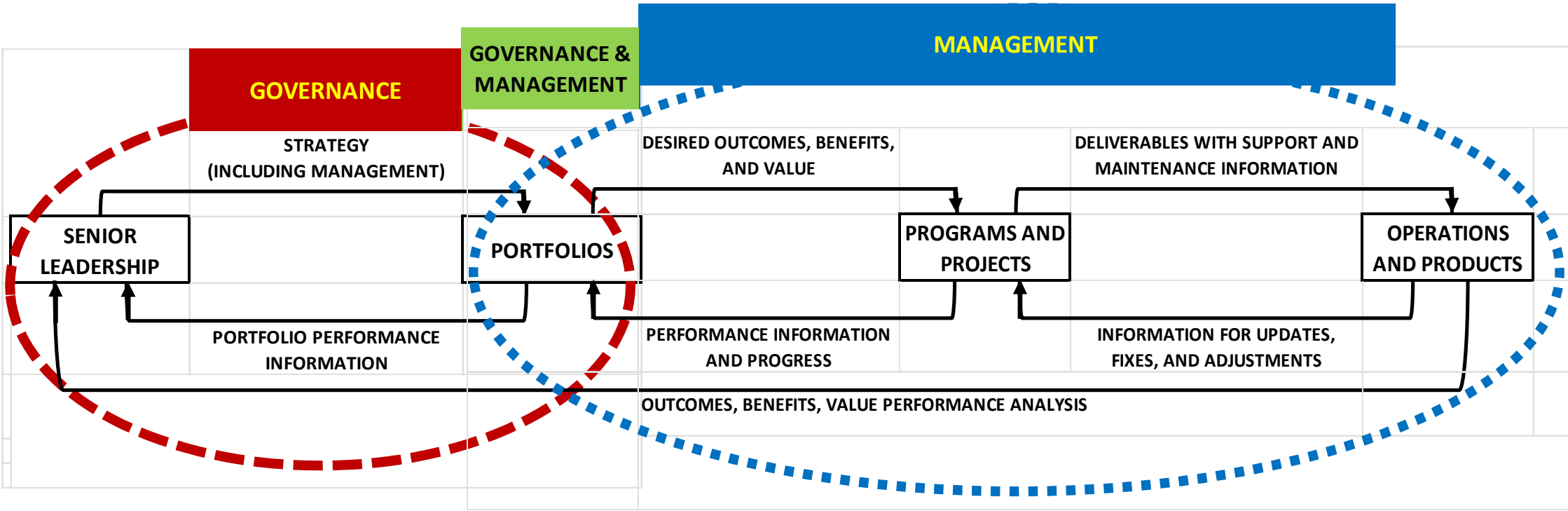
AS AN APPLICATION, IN SOLID WASTE PROJECT MANAGEMENT, SUSTAINABILITY, INNOVATION, PRIVACY AND BUSINESS ANALYSIS ADJUSTED TO RISK, WITH SPECIAL ATTENTION TO SAFETY, LABOR CONSTRAINTS, LOGISTICS, AND INSTITUTIONAL GOVERNANCE, ORIENT THE ROLE OF DIGITAL TWINS.




SYSTEM GOVERNANCE&MANAGEMENT
(ADAPTATION FIGURE 1-4, PMBOK7, P. 17)
SYSTEM´S ENVIRONMENT
GOVERNANCE
GOVERNANCE& MANAGEMENT
MANAGEMENT
INTRA SYSTEM ORGANIZATIONAL ENVIRONMENT
➔REAL-LIFE TWIN
IN-PORTFOLIO OPERATIONS&PRODUCTS
BUSINESS VALUE REALIZATION BY DISSEMINANTING AND IMPLEMENTING BETTER DECISION MAKING MANAGEMENT THROUGHOUT THE ORGANIZATION
PORTFOLIO REVIEW AND ADJUSTMENTS
➔PROJECT CHARTER
➔PROJECT MANAGEMENT PLAN
➔PROGRAM CHARTER
➔PROGRAM MANAGEMENT PLAN
PARTIAL DELIVERABLES
PARTIAL DELIVERIES
IN-PROJECT & IN-PROGRAM OPERATIONS & PRODUCTS
BUSINESS VALUE REALIZATION BY DELIVERING PARTIAL INCREMENTAL OPERATIONAL ITERATIVE RESULTS
VISION MISSION STRATEGY
PORTFOLIO
DECISIONS ON BUSINESS VALUE OBJECTIVES
PROGRAMS PROJECTS
RESULTS DELIVERY
OPERATIONS & PRODUCTS
BUSINESS VALUE REALIZATION
SYSTEM EXPLOITATION
IN OPERATION
DECLINE
RETIREMENT
BUSINESS IMPACT ANALYSIS
➔BUSINESS CASE
➔BENEFITS MANAGEMENT PLAN
➔PROJECT&PROGRAM SUCCESS MEASURES
VALUE PERFORMANCE ANALYSIS
➔BUSINESS CASE
➔BENEFITS MANAGEMENT PLAN
➔PROJECT&PROGRAM SUCCESS MEASURES
VALUE CREATION THROUGH AGILITY
GOVERNANCE & MANAGEMENT WORKFLOW
FEEDBACK



ADAPTATION FIG. 2.2-EXAMPLE OF A SYSTEM FOR VALUE DELIVERY (PMBOK8S, PP. 15)



ADAPTATION: FIG. 2.3-EXAMPLE INFORMATION FLOW (PMBOK8S, PP. 16)

CAPABILITY LOGIC→WHAT / AI AGENT LOGIC→HOW

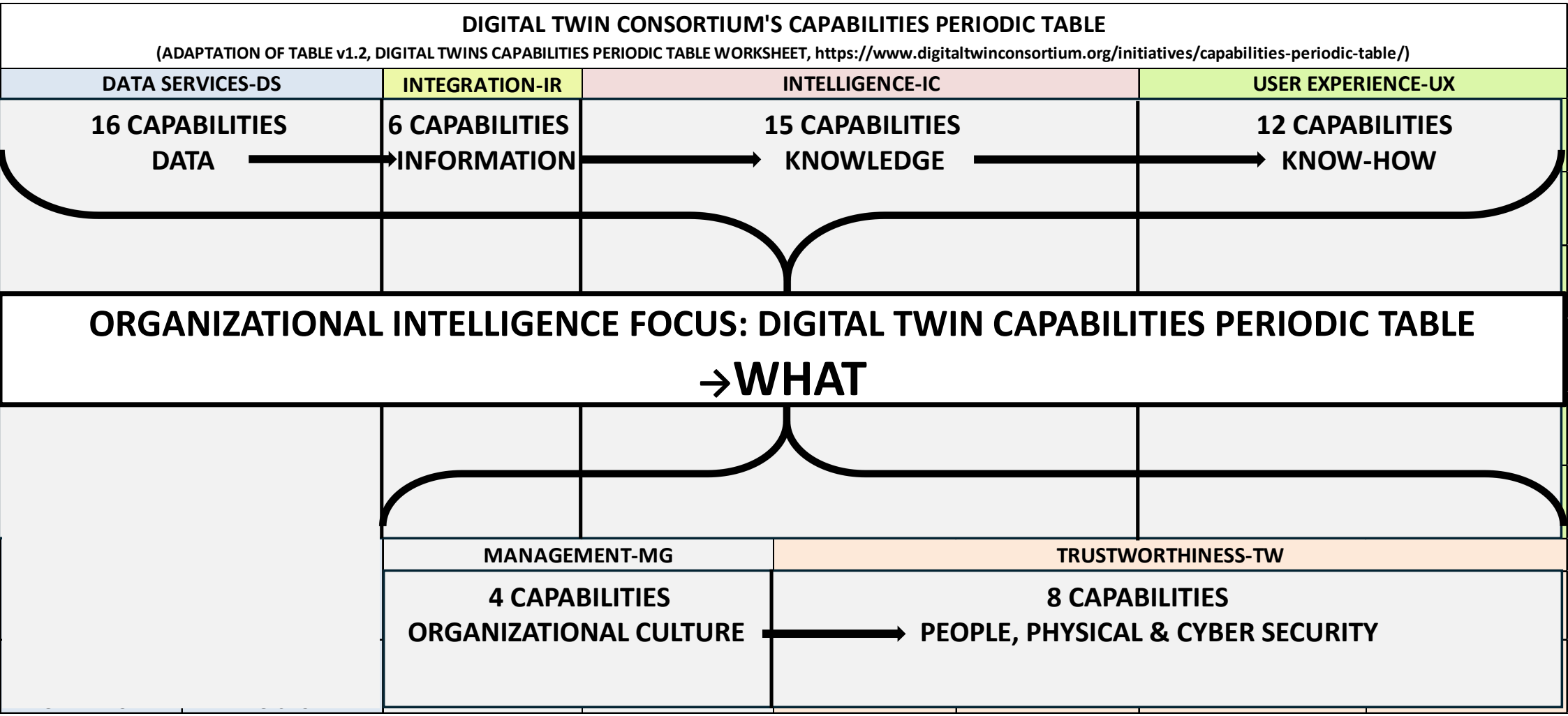
FIRST, STUDENTS DECIDE **WHAT** THE DIGITAL TWIN MUST PROVIDE USING THE **DIGITAL TWIN CONSORTIUM CAPABILITIES PERIODIC TABLE CATEGORIES**, WHICH ENCOMPASS DS (DATA SERVICES), IR (INTEGRATION), IC (INTELLIGENCE), UX (USER EXPERIENCE), MG (MANAGEMENT), AND TW (TRUSTWORTHINESS).

CONSISTENT WITH THE TABLE'S FRAMING, THESE CATEGORIES CAN BE READ AS A VALUE PROGRESSION FROM DATA TO INFORMATION, KNOWLEDGE, KNOW-HOW AND, ULTIMATELY, ORGANIZATIONAL INTELLIGENCE.

DIGITAL TWIN CONSORTIUM'S CAPABILITIES PERIODIC TABLE

(ADAPTATION OF TABLE v1.2, DIGITAL TWINS CAPABILITIES PERIODIC TABLE WORKSHEET, https://www.digitaltwinconsortium.org/initiatives/capabilities-periodic-table/)

DATA SERVICES-DS		INTEGRATION-IR	INTELLIGENCE-IC			USER EXPERIENCE-UX	
DS.AR AI MODEL REPOSITORY	DS.AG DATA AGGREGATION	IR.AS API SERVICES				UX.GM GAMIFICATION	UX.DB DASHBOARDS
DS.SR SIMULATION MODEL REPOSITORY	DS.AS ASYNCHRONOUS INTEGRATION	IR.CL COLLABORATION PLATFORM INTEGATION	IC.CS COMPOSITION	IC.SM SIMULATION	IC.RP REPORTING	UX.3R 3D RENDERING	UX.XR EXTENDED REALITY (XR)
DS.SA DATA STORAGE AND ARCHIVE SERVICES	DS.RT REAL-TIME PROCESSING	IR.DT DIGITAL TWIN INTEGRATION	IC.DL DISTRIBUTED LEDGER AND SMART CONTRACTS	IC.FL FEDERATED LEARNING	IC.AL ALERTS AND NOTIFICATION	UX.GE GAMING ENGINE VISUALIZATION	UX.ER ENTITY RELATIONSHIP VISUALIZATION
DS.DS DOMAIN SPECIFIC DATA MANAGEMENT	DS.BP BATCH PROCESSING	IR.IO OT/IoT SYSTEM INTEGRATION	IC.BR BUSINESS RULES	IC.AI ARTIFICIAL INTELLIGENCE	IC.OS ORCHESTRATION	UX.BP BUSINESS PROCESS MANAGEMENT&WORKFLOW	UX.RM REAL-TIME MONITORING
DS.IR DIGITAL TWIN INSTANCE REPOSITORY	DS.CX DATA CONTEXTUALIZATION	IR.EG ENGINEERING SYSTEMS INTEGRATION	IC.PS PRESCRIPTIVE RECOMMENDATIONS	IC.PR PREDICTION	IC.IC COMMAND AND CONTROL	UX.BI BUSINESS INTELLIGENCE	UX.AV ADVANCED VISUALIZATION
DS.RP DIGITAL TWIN MODEL REPOSITORY	DS.TR DATA TRANSFORMATION AND WRANGLING	IR.ET ENTERPRISE SYSTEM INTEGRATION	IC.MA MATHEMATICAL ANALYTICS	IC.AA DATA ANALYSIS AND ANALYTICS	IC.SR SEARCH	UX.CI CONTINUOUS INTELLIGENCE	UX.BV BASIC VISUALIZATION
DS.ON ONTOLOGY MANAGEMENT	DS.ST DATA STREAMING	MANAGEMENT-MG		TRUSTWORTHINESS-TW			
		MG.DG DATA GOVERNANCE	MG.SM SYSTEM MONITORING	TW.RP RESPONSIBILITY	TW.RL RELIABILITY	TW.PR PRIVACY	TW.DS DEVICE SECURITY
DS.SG SYNTHETIC DATA GENERATION	DS.AI DATA ACQUISITION AND INGESTION	MG.EL EVENT LOGGING	MG.DM DEVICE MANAGEMENT	TW.RS RESILIENCE	TW.SF SAFETY	TW.SC SECURITY	TW.EX DATA ENCRIPTION


DIGITAL TWIN CONSORTIUM'S CAPABILITIES PERIODIC TABLE
(ADAPTATION OF TABLE v1.2, DIGITAL TWINS CAPABILITIES PERIODIC TABLE WORKSHEET, https://www.digitaltwinconsortium.org/initiatives/capabilities-periodic-table/)
DATA SERVICES-DS
INTEGRATION-IR
INTELLIGENCE-IC
USER EXPERIENCE-UX
16 CAPABILITIES
DATA
6 CAPABILITIES
INFORMATION
15 CAPABILITIES
KNOWLEDGE
12 CAPABILITIES
KNOW-HOW
ORGANIZATIONAL INTELLIGENCE FOCUS: DIGITAL TWIN CAPABILITIES PERIODIC TABLE
→WHAT
MANAGEMENT-MG
TRUSTWORTHINESS-TW
4 CAPABILITIES
ORGANIZATIONAL CULTURE
8 CAPABILITIES
PEOPLE, PHYSICAL & CYBER SECURITY

CAPABILITY LOGIC→WHAT / AI AGENT LOGIC→HOW

SECOND, STUDENTS DECIDE HOW SELECTED FUNCTIONS WILL BE PERFORMED USING THE AI AGENT CAPABILITIES PERIODIC TABLE CATEGORIES IN THE SAME FORMAT: PW (PERCEPTION & KNOWLEDGE), CG (COGNITION & REASONING), LA (LEARNING & ADAPTATION), AE (ACTION & EXECUTION), IC (INTERACTION & COLLABORATION), AND GS (GOVERNANCE & SAFETY).

THIS SEPARATION IS ESPECIALLY IMPORTANT IN EDUCATIONAL SETTINGS WHERE GENERATIVE AI CAN BIAS TEAMS TOWARD SOLUTIONS THAT SOUND POWERFUL BUT HAVE UNCLEAR REQUIREMENTS, WEAK GOVERNANCE, OR POOR EXECUTION REALISM.

IT IS WORTHWHILE TO EMPHASIZE THE ROLE OF THE SYSTEM PERFORMANCE MATRIX (EXCELLENCE LEVEL AGAINST READINESS DEGREES), IN THE CONTEXT OF DTC'S AI AGENT CAPABILITIES PERIDIC TABLE, IN TERMS OF 11 CAPABILITIES OF ORGANIZATIONAL INTELLIGENCE AGAINST 12 CAPABILITIES IC-INTERACTION & COLLABORATIONS.

DIGITAL TWIN CONSORTIUM-DTC: AI AGENT CAPABILITIES PERIODIC TABLE

DIGITAL TWIN CONSORTIUM'S AI AGENT CAPABILITIES PERIODIC TABLE (AIA CPT)

(ADAPTATION OF TABLE v1.0 USER GUIDE, AI AGENT CAPABILITIES PERIODIC TABLE INTERACTIVE FRAMEWORK AND TOOLKIT, https://www.digitaltwinconsortium.org/initiatives/ai-agent-capabilities-periodic-table/)

PK-PERCEPTION & KNOWLEDGE	CG-COGNITION & REASONING	LA-LEARNING & ADAPTATION	AE-ACTION & EXECUTION	IC-INTERACTION & COLLABORATION	
	CG.PL PLANNING & DECOMPOSITION	LA.MM MEMORY MANAGEMENT		IC.NL NATURAL LANGUAGE	IC.CS CONSENSUS PROTOCOLS
	CG.RS REASONING	LA.RL REINFORCEMENT LEARNING	AE.TX TASK EXECUTION & IMPLEMENTATION	IC.DM DIALOGUE MANAGEMENT	IC.CF CONFLICT RESOLUTION
PK.OB ENVIRONMENTAL SENSING	CG.DC DECISION MAKING	LA.AD SELF-OPMIZATION	AE.TL TOOL USAGE & API INTEGRATION	IC.HL HUMAN-IN-LOOP	IC.SI INDUSTRIAL INTEGRATION
PK.KB KNOWLEDGE ACCESS	CG.PS PROBLEM SOLVING	LA.SL SUPERVISED LEARNING	AE.CG CODE GENERATION & EXECUTION	IC.AC AGENT COMMUNICATION	IC.ES ENTERPRISE INTEGRATION
PK.CX CONTEXT & MEMORY	CG.PP FORMAL PLANNING	LA.VM VECTOR MEMORY	AE.CX CONTENT CREATION & GENERATION	IC.CL COLLABORATION	IC.MB MESSAGE BROKERS
PK.MF MULTI-MODAL FUSION	CG.PA PLAN ADAPTATION	LA.MS MEMORY SCORING	AE.TM TOOL LIFECYCLE MANAGEMENT	IC.RB ROLE BEHAVIOR	IC.DS DISTRIBUTED COORDINATION
GS-GOVERNANCE & SAFETY			AE.MC MCP INTEGRATION	GS-GOVERNANCE & SAFETY	
GS.DL DEPLOYMENT MANAGEMENT	GS.MO MONITORING	GS.EV EVALUATION	GS-GOVERNANCE & SAFETY	GS.ET ETHICS	GS.PR PRIVACY
GS.SC SCALING	GS.SF SAFETY	GS.SE SECURITY	GS.EX EXPLANABILITY	GS.RL RELIABILITY	GS.TC TRUST MANAGEMENT

DIGITAL TWIN CONSORTIUM'S AGENT CAPABILITIES PERIODIC TABLE (AIA CPT)

(ADAPTATION OF TABLE v1.0 USER GUIDE, AI AGENT CAPABILITIES PERIODIC TABLE INTERACTIVE FRAMEWORK AND TOOLKIT, https://www.digitaltwinconsortium.org/initiatives/ai-agent-capabilities-periodic-table/)

PK-PERCEPTION & KNOWLEDGE	CG-COGNITION & REASONING	LA-LEARNING & ADAPTATION	AE-ACTION & EXECUTION		IC-INTERACTION & COLLABORATION
4 CAPABILITIES DATA	6 CAPABILITIES INFORMATION	6 CAPABILITIES KNOWLEDGE	6 CAPABILITIES KNOW-HOW	SYSTEM PERFORMANCE MATRIX READINESS x EXCELLENCE	12 CAPABILITIES (EVs) ORGANIZATIONAL READINESS: >PDCA: EXISTENCE >LIFECYCLE: EXPERIENCE >NONAKA: SPIRAL OF KNOWLEDGE CREATION & TRANSFERING QUINN'S COMPETING VALUES: EMERGENCE >BEGHTOL: EVIDENCE

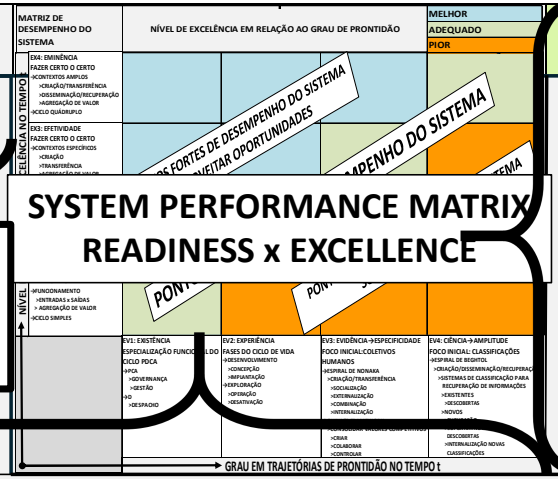
11 CAPABILITIES
ORGANIZATIONAL INTELLIGENCE EXCELLENCE (EXs): EFFICIENCY/EFFICACY/EMINENCE/EFFECTIVENESS

GS-GOVERNANCE & SAFETY		GS-GOVERNANCE & SAFETY	
	GS-GOVERNANCE & SAFETY		

DIGITAL TWIN CONSORTIUM'S AGENT CAPABILITIES PERIODIC TABLE (AIA CPT)

(ADAPTATION OF TABLE v1.0 USER GUIDE, AI AGENT CAPABILITIES PERIODIC TABLE INTERACTIVE FRAMEWORK AND TOOLKIT, https://www.digitaltwinconsortium.org/initiatives/ai-agent-capabilities-periodic-table/)

PK-PERCEPTION & KNOWLEDGE	CG-COGNITION & REASONING	LA-LEARNING & ADAPTATION	AE-ACTION & EXECUTION	IC-INTERACTION & COLLABORATION
4 CAPABILITIES DATA →	6 CAPABILITIES INFORMATION →	6 CAPABILITIES KNOWLEDGE →	6 CAPABILITIES KNOW-HOW	12 CAPABILITIES (EVs) ORGANIZATIONAL READINESS: >PDCA: EXISTENCE >LIFECYCLE: EXPERIENCE >NONAKA: SPIRAL OF KNOWLEDGE CREATION & TRANSFERING QUINN'S COMPETING VALUES: EMERGENCE >BEGHTOL: EVIDENCE



ORGANIZATIONAL INTELLIGENCE FOCUS: AI AGENTE DIGITAL TWIN CAPABILITIES PERIODIC TABLE →HOW

11 CAPABILITIES
ORGANIZATIONAL INTELLIGENCE EXCELLENCE (EXs): EFFICIENCY/EFFICACY/EMINENCE/EFFECTIVENESS

GS-GOVERNANCE & SAFETY	GS-GOVERNANCE & SAFETY	GS-GOVERNANCE & SAFETY

TAKING INTO ACCOUNT THE DEGREE OF EMBEDDED TECHNOLOGY, IN ADDITION TO STATIC AUTOMATION (TYPE 0), DTC'S AIACPT CONSIDERS 4 TYPES OF MORE SOPHISTICATED AI AGENTS:

0- STATIC AUTOMATION: PRE-PROGRAMMED RESPONSES WITH NO LEARNING OR ADAPTATION & TRADITIONAL RULE-BASED SYMBOLIC AI SYSTEMS AND BASIC AUTOMATION.

1- CONVERSATIONAL AGENTS: NATURAL LANGUAGE INTERACTION WITH BASIC CONTEXT MANAGEMENT & CUSTOMER SERVICE CHATBOTS, FAQ SYSTEMS, VOICE ASSISTANT.

2- PROCEDURAL WORKFLOW AGENTS: MULTI-STEP TASK EXECUTION WITH TOOL INTEGRATION AND ROLE-BASED COLLABORATION & WORKFLOW AUTOMATION, TASK ORCHESTRATION, PROCESS MANAGEMENT.

3- COGNITIVE AUTONOMOUS AGENTS: SELF-DIRECTED PLANNING, LEARNING FROM EXPERIENCE, SOPHISTICATED REASONING & RESEARCH ASSISTANTS, PREDICTIVE SYSTEMS, AUTONOMOUS DECISION-MAKING.

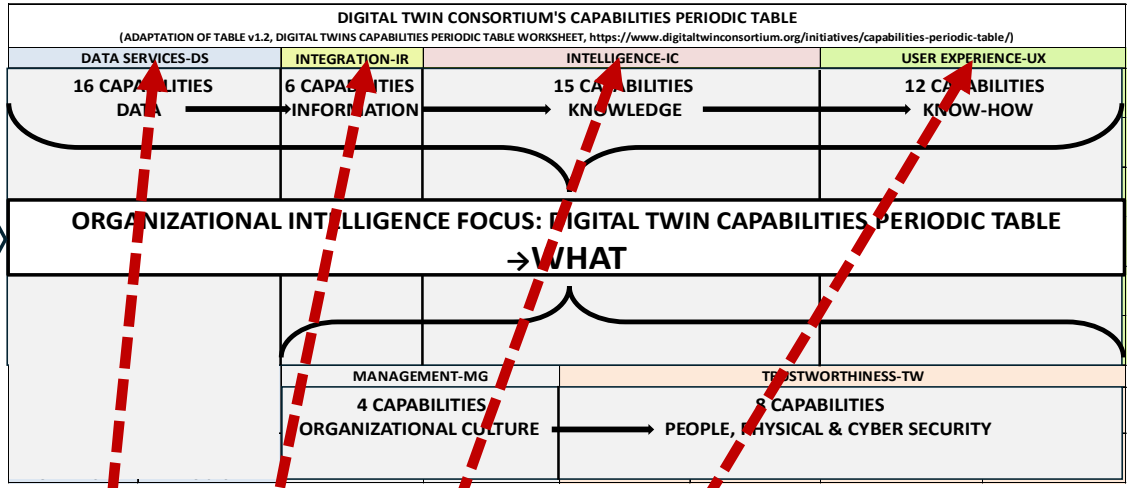
4- COLLABORATIVE INTELLIGENCE AGENTS: COMPLEX ORCHESTRATION, SUCH AS SMART CITY SYSTEMS.

CONSOLIDATION OF THE TWO-LAYER LOGIC USED IN THE INSTRUCTIONAL PATTERN.

AT THE WHAT (TWIN) LAYER, TEAMS MAKE A REQUIREMENTS-ORIENTED SELECTION OF DIGITAL TWIN CAPABILITY CATEGORIES (DS/IR/IC/UX/MG/TW) TO JUSTIFY WHAT THE DSQ MUST PROVIDE TO SUPPORT DECISIONS.

AT THE HOW (TWIN/AGENT) LAYER, TEAMS MAKE A REQUIREMENTS-ORIENTED SELECTION OF AI-AGENT CAPABILITY CATEGORIES (PW/CG/LA/AE/IC/GS) TO JUSTIFY HOW SELECTED DSQ FUNCTIONS WILL BE IMPLEMENTED TO SUPPORT DECISIONS, WITH THE CHOICE OF AGENT MODALITY (CONVERSATIONAL, WORKFLOW, AUTONOMOUS, OR COLLABORATIVE) EXPLICITLY CONSTRAINED BY GOVERNANCE AND SAFETY REQUIREMENTS.

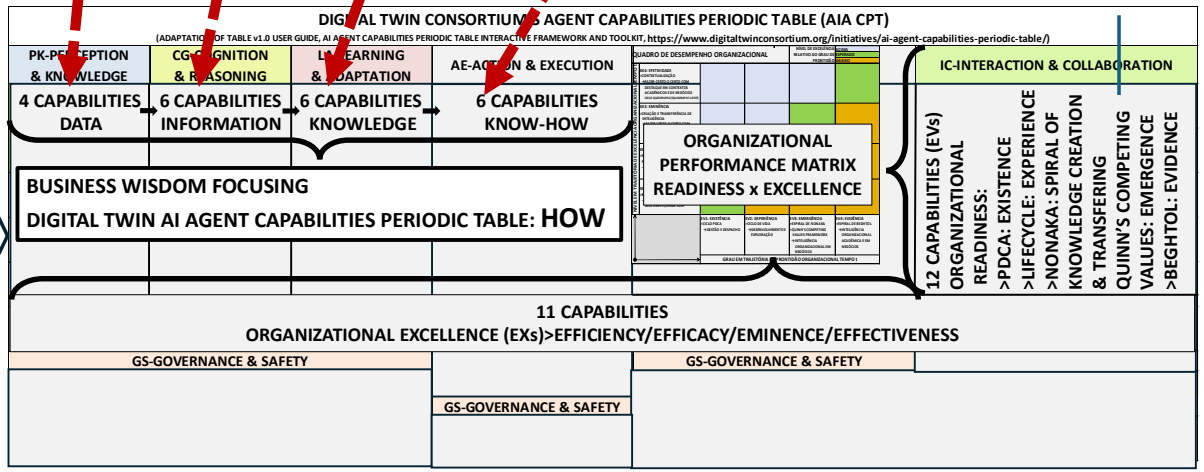
DIGITAL TWIN CAPABILITIES PERIODIC TABLE → **WHAT**



DIGITAL TWIN AI AGENT TYPES → **HOW**

1. **CONVERSATIONAL AGENTS**: NATURAL LANGUAGE INTERACTION WITH BASIC CONTEXT MANAGEMENT & CUSTOMER SERVICE CHATBOTS, FAQ SYSTEMS, VOICE ASSISTANT
2. **PROCEDURAL WORKFLOW AGENTS**: MULTI-STEP TASK EXECUTION WITH TOOL INTEGRATION AND ROLE-BASED COLLABORATION & WORKFLOW AUTOMATION, TASK ORCHESTRATION, PROCESS MANAGEMENT
3. **COGNITIVE AUTONOMOUS AGENTS**: SELF-DIRECTED PLANNING, LEARNING FROM EXPERIENCE, SOPHISTICATED REASONING & RESEARCH ASSISTANTS, PREDICTIVE SYSTEMS, AUTONOMOUS DECISION-MAKING
4. **COLLABORATIVE INTELLIGENCE WITH EMERGENT BEHAVIORS AND DISTRIBUTED COORDINATION & INDUSTRIAL MULTI-AGENT GENERATIVE SYSTEMS (MAGS)**: AUTOMATION, SMART CITY SYSTEMS, COMPLEX ORCHESTRATION

DIGITAL TWIN AI AGENT CAPABILITIES PERIODIC TABLE → **HOW**



IT/OT/ET CONVERGENCE AND WHY IT IS CENTRAL TO "DIGITAL TWIN-FIRST"

DTs BECOME MOST CHALLENGING AND MOST VALUABLE PRECISELY WHEN THEY SIT AT THE INTERSECTION OF IT (INFORMATION TECHNOLOGY SYSTEMS AND DATA/ANALYTICS), OT (OPERATIONAL TECHNOLOGY SUCH AS SENSORS, AUTOMATION, AND INDUSTRIAL CONTROL), AND ET (ENGINEERING TECHNOLOGY SUCH AS MODELS, DESIGN INTENT, AND SYSTEM ENGINEERING ARTIFACTS).

IN PRACTICE, DT MUST RECONCILE HETEROGENEOUS DATA SOURCES AND TIME SCALES:

- **IT OFTEN OPERATES ON BUSINESS RHYTHMS AND TRANSACTIONAL CORRECTNESS;**
- **OT OPERATES ON REAL-TIME CONSTRAINTS, SAFETY, AND RELIABILITY; AND**

ET OPERATES ON DESIGN FIDELITY, PHYSICS, AND LIFECYCLE REASONING.

WHEN THESE WORLDS CONVERGE, REQUIREMENTS ALSO CONVERGE:

- **INTEGRATION AND DATA SERVICES CANNOT BE SEPARATED FROM TRUSTWORTHINESS, AND INTELLIGENCE CANNOT BE SEPARATED FROM GOVERNANCE AND OPERATIONAL SAFETY; AND**
- **THE INSTRUCTIONAL DESIGN LEVERAGES THIS CONVERGENCE TO MAKE STUDENTS PRACTICE REALISTIC REASONING.**

REAL-LIFE SYSTEM-IN-QUESTION (RSQ) AND THE CORRESPONDING DIGITAL SYSTEM-IN-QUESTION (DSQ) BOUNDARIES MUST EXPLICITLY INCLUDE WHERE IT ENDS AND OT BEGINS (E.G., WHAT DATA IS AVAILABLE, AT WHAT CADENCE, UNDER WHAT SECURITY CONSTRAINTS) AND HOW ET REPRESENTATIONS ARE MAINTAINED (E.G., WHETHER THE DSQ INCLUDES ENGINEERING MODELS, ASSUMPTIONS, AND THEIR VERSIONING).

FROM DIGITAL REPLICA TO DECISION REFERENCE

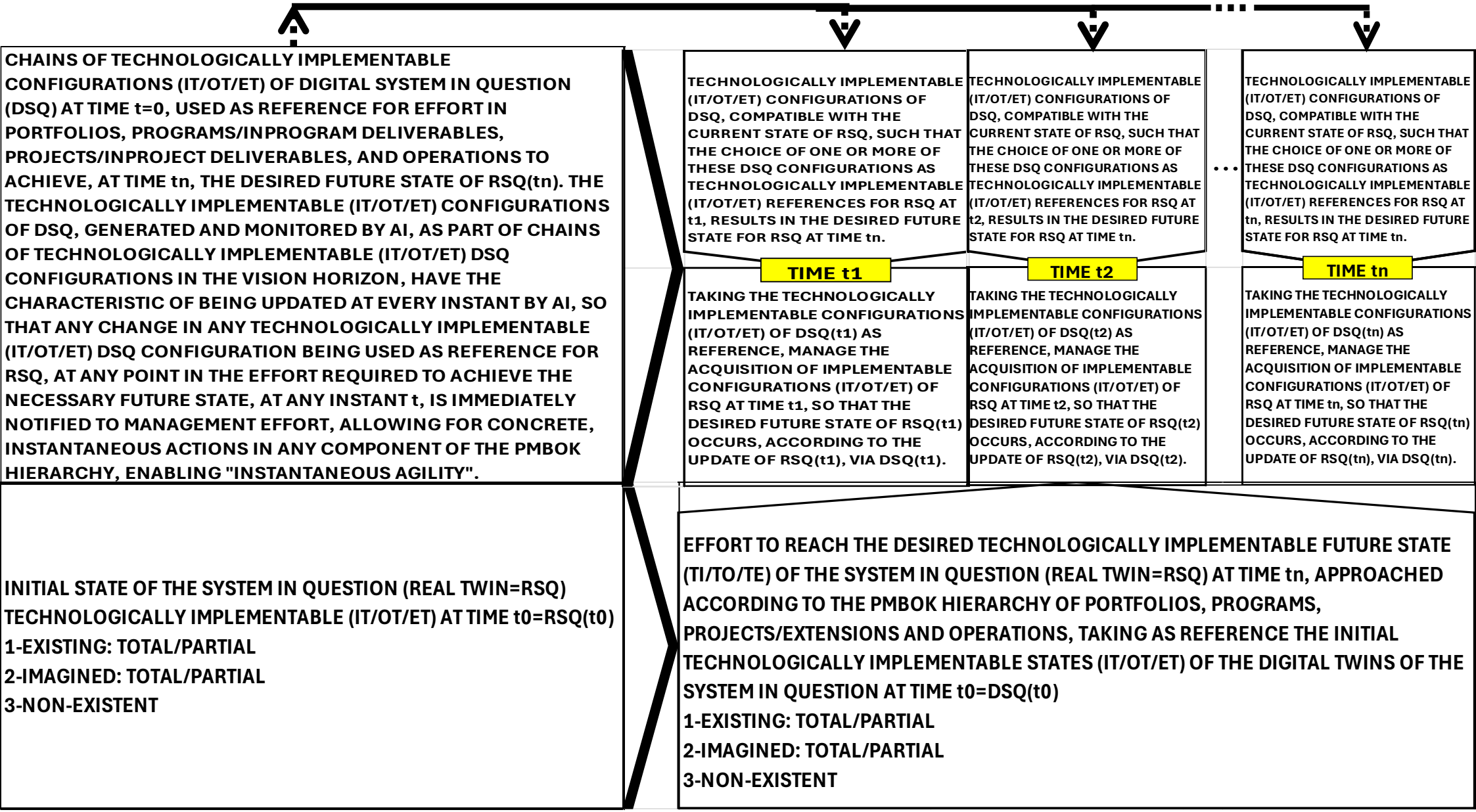
A DT BECOMES STRATEGICALLY MEANINGFUL WHEN IT HELPS DECISION-MAKERS CHOOSE AMONG ALTERNATIVES AND JUSTIFY CHANGE SEQUENCING.

WHEN STUDENTS TREAT DT AS A REPLICA, THEIR WORK TENDS TO OPTIMIZE FIDELITY AGAINST TODAY'S CONDITIONS.

WHEN THEY TREAT DT AS A REFERENCE, THEIR WORK TENDS TO OPTIMIZE DECISION DEFENSIBILITY TOWARD A FUTURE TARGET.

THIS MATTERS BECAUSE REAL SYSTEMS ARE NOT STATIC: CONSTRAINTS CHANGE, OPERATIONS EVOLVE, RISKS MATERIALIZE, AND THE INFORMATION ENVIRONMENT SHIFTS.

TREATING THE CORRESPONDING DT AS A REFERENCE FORCES STUDENTS TO ARTICULATE A DESIRED FUTURE STATE AND MAKE THE PATHWAY EXPLICIT, RATHER THAN IMPLICITLY ASSUMING THAT IMPROVEMENT IS THE NATURAL CONSEQUENCE OF MEASUREMENT.


CHAINS OF TECHNOLOGICALLY IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF DIGITAL SYSTEM IN QUESTION (DSQ) AT TIME t=0, USED AS REFERENCE FOR EFFORT IN PORTFOLIOS, PROGRAMS/INPROGRAM DELIVERABLES, PROJECTS/INPROJECT DELIVERABLES, AND OPERATIONS TO ACHIEVE, AT TIME tn, THE DESIRED FUTURE STATE OF RSQ(tn). THE TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) CONFIGURATIONS OF DSQ, GENERATED AND MONITORED BY AI, AS PART OF CHAINS OF TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) DSQ CONFIGURATIONS IN THE VISION HORIZON, HAVE THE CHARACTERISTIC OF BEING UPDATED AT EVERY INSTANT BY AI, SO THAT ANY CHANGE IN ANY TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) DSQ CONFIGURATION BEING USED AS REFERENCE FOR RSQ, AT ANY POINT IN THE EFFORT REQUIRED TO ACHIEVE THE NECESSARY FUTURE STATE, AT ANY INSTANT t, IS IMMEDIATELY NOTIFIED TO MANAGEMENT EFFORT, ALLOWING FOR CONCRETE, INSTANTANEOUS ACTIONS IN ANY COMPONENT OF THE PMBOK HIERARCHY, ENABLING "INSTANTANEOUS AGILITY".
TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) CONFIGURATIONS OF DSQ, COMPATIBLE WITH THE CURRENT STATE OF RSQ, SUCH THAT THE CHOICE OF ONE OR MORE OF THESE DSQ CONFIGURATIONS AS TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) REFERENCES FOR RSQ AT t1, RESULTS IN THE DESIRED FUTURE STATE FOR RSQ AT TIME tn.
TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) CONFIGURATIONS OF DSQ, COMPATIBLE WITH THE CURRENT STATE OF RSQ, SUCH THAT THE CHOICE OF ONE OR MORE OF THESE DSQ CONFIGURATIONS AS TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) REFERENCES FOR RSQ AT t2, RESULTS IN THE DESIRED FUTURE STATE FOR RSQ AT TIME tn.
TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) CONFIGURATIONS OF DSQ, COMPATIBLE WITH THE CURRENT STATE OF RSQ, SUCH THAT THE CHOICE OF ONE OR MORE OF THESE DSQ CONFIGURATIONS AS TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) REFERENCES FOR RSQ AT tn, RESULTS IN THE DESIRED FUTURE STATE FOR RSQ AT TIME tn.
TIME t1
TIME t2
TIME tn
TAKING THE TECHNOLOGICALLY IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF DSQ(t1) AS REFERENCE, MANAGE THE ACQUISITION OF IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF RSQ AT TIME t1, SO THAT THE DESIRED FUTURE STATE OF RSQ(t1) OCCURS, ACCORDING TO THE UPDATE OF RSQ(t1), VIA DSQ(t1).
TAKING THE TECHNOLOGICALLY IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF DSQ(t2) AS REFERENCE, MANAGE THE ACQUISITION OF IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF RSQ AT TIME t2, SO THAT THE DESIRED FUTURE STATE OF RSQ(t2) OCCURS, ACCORDING TO THE UPDATE OF RSQ(t2), VIA DSQ(t2).
TAKING THE TECHNOLOGICALLY IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF DSQ(tn) AS REFERENCE, MANAGE THE ACQUISITION OF IMPLEMENTABLE CONFIGURATIONS (IT/OT/ET) OF RSQ AT TIME tn, SO THAT THE DESIRED FUTURE STATE OF RSQ(tn) OCCURS, ACCORDING TO THE UPDATE OF RSQ(tn), VIA DSQ(tn).
INITIAL STATE OF THE SYSTEM IN QUESTION (REAL TWIN=RSQ) TECHNOLOGICALLY IMPLEMENTABLE (IT/OT/ET) AT TIME t0=RSQ(t0)
1-EXISTING: TOTAL/PARTIAL
2-IMAGINED: TOTAL/PARTIAL
3-NON-EXISTENT
EFFORT TO REACH THE DESIRED TECHNOLOGICALLY IMPLEMENTABLE FUTURE STATE (TI/TO/TE) OF THE SYSTEM IN QUESTION (REAL TWIN=RSQ) AT TIME tn, APPROACHED ACCORDING TO THE PMBOK HIERARCHY OF PORTFOLIOS, PROGRAMS, PROJECTS/EXTENSIONS AND OPERATIONS, TAKING AS REFERENCE THE INITIAL TECHNOLOGICALLY IMPLEMENTABLE STATES (IT/OT/ET) OF THE DIGITAL TWINS OF THE SYSTEM IN QUESTION AT TIME t0=DSQ(t0)
1-EXISTING: TOTAL/PARTIAL
2-IMAGINED: TOTAL/PARTIAL
3-NON-EXISTENT